

ITAI 3375 — Module 13

Lab 13: Data Economics & Career Reflection

Connecting your curation skills to the job market and your own path **SLOs:** 1, 5 **Points:** 100

Overview

This lab has two parts that work together. In Part 1 you will put a dollar value on the curation work you did in Lab 10 — the same kind of business case you would make to justify a data quality project at a real organisation. In Part 2 you will look at where data curation roles are available right now and build a short, concrete plan for the next 90 days of your career development.

Everything in this lab is written work. There is no code. There are no datasets to load. The hardest part is thinking clearly — and writing honestly.

This is personal. Part 2 asks you to research real job postings and write a real career plan. The more specific you are about your own situation, goals, and constraints as an evening student, the more useful this lab will be to you.

Part 1 — Put a Dollar Value on Your Lab 10 Curation Work (Evening 1, ~60 minutes)

In Module 13 we looked at how organisations calculate the ROI of data curation investments. Now you will apply that thinking to your own Lab 10 project.

You do not need real financial data or precise numbers. This is an estimation exercise. The goal is to practice the reasoning — identifying what got better, what that is worth, and how to communicate it in business language.

1a. Describe what you improved

Go back to your Lab 10 quality scores (Step 4) and your curation notes (Step 2). For each change you made to the dataset, write one sentence describing:

- What was the problem before? (e.g., '18% of records had a missing date field')
- What did you do? (e.g., 'Removed 94 rows where the date was missing')
- What is the situation after? (e.g., 'Date field is now 100% complete')

List at least two improvements. These can be small — fixing a date format counts, removing duplicates counts, standardising a category column counts.

1. Missing values in key text/category columns

Before: There were 26,114 records lacking a descriptor, 57,907 missing a city, and 8,704 with a missing incident ZIP.

What I did: I populated the missing values in the descriptor, city, and incident_zip columns with the term "Unknown."

After: These three fields are now fully complete (no blank or null values exist).

2. Inconsistent date formats

Before: The created_date and closed_date columns were recorded as text/object values with varying formats, complicating analysis.

What I did: I transformed both columns into the correct datetime format using `pd.to\_datetime` (with errors coerced to NaT when necessary).

After: Both date fields are now uniform datetime values that can be utilized directly for time-based analysis.

1b. Estimate the business value

For each improvement you listed, pick the category below that best describes its value and write a oneparagraph estimate.

You are not expected to produce precise numbers — you are expected to reason through the logic.

Value Category	How to Estimate It
Time saved	If an analyst downstream did not have to fix this themselves, how much time did you save? Estimate in hours. At a rough analyst rate of \$35–50/hour, what is that worth? (Example: 'Without this fix, an analyst would spend roughly 2 hours untangling the date formats. At \$40/hour that is \$80 of saved analyst time.') Prior to the conversion, any analyst aiming to compute response times, analyze monthly trends, or filter recent requests would have needed to manually re-parse the text/object date columns. For a dataset containing 1 million rows, this is not a simple one-time task; it generally involves writing and debugging conversion code, addressing edge cases, and re-validating the outcomes. I estimate that this process would consume approximately 1.5 to 2.5 hours of a data analyst's time each time the raw file is utilized. At a conservative internal rate of \$40 to \$45 per hour, this translates to a savings of \$60 to \$110 in analyst time for each analysis cycle. Following the standardization, the dates can be used immediately for time-based calculations without any additional cleaning required.
Error risk reduced	If the bad data went into a model, what kind of wrong output could it cause? How often? What is one such error worth to a business — a bad recommendation, a wrong decision, a compliance flag? (Example: 'Missing income fields could cause the recommendation engine to default to the wrong product tier for up to 18% of users.')

	Improvement Filling in the missing values for descriptors, city, and incident_zip with "Unknown" Category: Reduced error risk / Enhanced model quality Without this adjustment, a classification model designed to predict the urgency of service requests would encounter incomplete input features for over 26,000 rows missing descriptors, more than 57,000 missing cities, and around 8,000 missing ZIP codes. This lack of information could lead the model to incorrectly classify urgency levels, particularly between "Needs review" and "High urgency" cases, or it might necessitate the exclusion of those records altogether. In the context of city services, even a slight increase of 5–10% in misdirected high-urgency complaints related to heating, plumbing, elevator, or traffic-signal issues could result in delayed responses and incur significant operational or safety costs. By clearly designating these gaps as "Unknown," the refined data mitigates that risk and provides any subsequent model with cleaner, more uniform features for learning.
Compliance risk reduced	If the data had a privacy or quality issue that could attract a regulatory penalty, what is the rough exposure? GDPR fines can reach €20M or 4% of global revenue. Even a small dataset with poorly handled personal data carries non-trivial risk. (Example: 'The unmasked email addresses in the raw file represent potential GDPR exposure. Removing them eliminated that risk.')
Model quality improved	If a model trained on your cleaned data is more accurate than one trained on the raw data, estimate the improvement. Module 13 examples showed that cleaning can improve model accuracy by 3–15%. What would a 5% accuracy improvement mean in the context of your dataset's intended AI use case?

Honest estimates beat invented precision: Writing 'I estimate this saves approximately 1–2 hours of analyst time per month, worth \$40–80, because the date normalisation currently requires manual correction before any analysis can run' earns full credit. Fabricating a specific number without reasoning does not.

1c. Write the business case

Combine your improvement descriptions and value estimates into a single short paragraph (150–250 words) that makes the case for the curation work you did. Write it as if you are presenting to a manager who does not know what pandas is — use plain language, focus on outcomes, and mention at least one dollar figure or time estimate.

Format: 'The curation work on [dataset name] addressed three issues that would otherwise have required... This saves approximately... and reduces the risk of...'

1c. Develop the business case

The curation efforts on the NYC 311 Service Requests dataset tackled two significant data-quality challenges that could have hindered analysis and compromised any models derived from the data. Firstly, numerous records were lacking critical information such as the complaint descriptor, city, and ZIP code; these deficiencies were clearly labeled as 'Unknown,' eliminating any ambiguity or blank entries. Secondly, the date fields indicating when each request was opened and closed were recorded in inconsistent text formats, which were then standardized into a uniform date-and-time format that any analyst can readily utilize.

In the absence of these improvements, an analyst would generally invest 1.5 to 2.5 hours each time the raw file is accessed merely to clean the dates and address the missing fields, resulting in a cost of approximately \$60 to \$110 at a standard internal rate of \$40 to \$45 per hour. The refined data also minimizes the likelihood that a model predicting request urgency would misclassify high-priority cases (such as heat, plumbing, elevator, or traffic-signal issues) due to incomplete or unusable data. In summary, the curation process prepares the data for both regular reporting and AI classification, while simultaneously conserving analyst time and reducing the risk of erroneous service prioritization.

What to submit for Part 1

- Your improvement descriptions from 1a (at least two).
- Your value estimates from 1b — one paragraph per improvement.
- Your business case paragraph from 1c.

Part 2 — Research the Job Market and Build a 90-Day Plan (Evening 1–2, ~60 minutes)

Data curation is a real career with real job openings. In this part you will look at what is actually available right now, assess where you fit, and write a specific 90-day action plan for your next steps.

2a. Find three real job postings

Search LinkedIn, Indeed, or your preferred job board for roles related to data curation, data quality, data engineering, or data governance. You are looking for roles you could realistically target within 6–18 months of completing this program — not necessarily right now.

For each of the three postings, fill in this table:

Field	Fill in for each of your three postings
Job Title	Data Quality analyst
Company & Location	CAPTRUST – Raleigh, NC hybrid

Salary Range (if listed)	mid-level range for this title is roughly \$70k–\$95k
Top 3 Required Skills	☐ Strong SQL + Python ☐ Data modeling and analysis concepts ☐ Stakeholder management / collaboration with data stewards and product managers
One Requirement You Already Meet	Through this course, I gained experience in cleaning and documenting actual datasets during Lab 10, where I worked on the NYC 311 dataset. This involved filling in missing values, standardizing dates, assessing data quality, and creating a Data Card
One Requirement You Do Not Yet Have	<i>(be honest)</i> Hands-on experience writing automated data-quality tests in dbt or embedding tests in a production analytics warehouse
Why This Role Appeals to You	<i>One</i> <i>This course is specifically designed to teach the essential skills needed for establishing and upholding data quality rules.sentence.</i>

2b. Identify your skills gap

Look across all three job postings. Write a short paragraph (100–150 words) answering:

- What skill or credential appears in all three (or two out of three) postings that you do not yet have?
- Is this a technical skill (SQL, Python, Spark, dbt), a tool (specific platform), a certification (AWS, CDMP), or a domain skill (healthcare data, financial data)?
- How significant is this gap — could you close it in a few weeks of focused learning, or does it require months of study?

2b. Recognizing Your Skills Gap

In reviewing the three job postings, the skill that stands out most prominently in two of them, and is strongly suggested in the third, is SQL paired with automated data-quality testing, particularly utilizing tools like dbt. This is a technical competency rather than merely a certification or domain expertise. Although I possess substantial experience with Python and pandas from my work on cleaning and analyzing the NYC 311 dataset in Lab 10, I have yet to create production-level SQL queries or develop automated quality tests that operate within a data warehouse.

The skills gap is moderate: the fundamental SQL principles and introductory dbt testing can be acquired through a few weeks of dedicated practice using free resources such as Kaggle, Mode Analytics SQL tutorials, or dbt's complimentary courses. Achieving the proficiency expected for mid-level positions will likely require about 2 to 3 months of regular evening practice, but this is attainable for someone studying part-time and does not necessitate years of additional formal education.

2c. Write your 90-day action plan

A 90-day plan has three milestones, one per month. Each milestone should be specific and achievable for someone working full-time and studying in the evenings. Vague milestones like 'learn more Python' do not count — write something measurable.

Month	What You Will Do
Month 1 — Build something	Complete one concrete project. This could be: add a second dataset to your Lab 10 Data Card, expand your annotation work, publish your Lab 10 curation notebook on GitHub, or complete one free tutorial (Kaggle Learn, DataCamp free tier, YouTube course). Describe the specific project and what you will have to show at the end of Month 1. I will publish my finalized Lab 10 NYC 311 curation notebook, which includes the cleaning steps, quality scores, a 50-row annotation, and the complete Data Card, to a public GitHub repository. By the end of Month 1, I will have a live GitHub link showcasing my actual data-curation work that I can share on LinkedIn and in job applications.
Month 2 — Learn one skill	Address the skills gap you identified in 2b. Pick one specific resource: a Coursera module, a HuggingFace tutorial, a YouTube playlist, or a free chapter of a book. Write down the resource name, the URL or title, and the specific skill you will have demonstrated at the end of Month 2. Complete the free “Introduction to SQL” course on Kaggle Learn and the free dbt Fundamentals course on the dbt Learn platform. By the end of Month 2 I will be able to write basic SQL queries (SELECT, JOIN, WHERE, GROUP BY) and create simple automated data-quality tests in dbt that check for nulls and uniqueness directly addressing the skills gap identified in 2b.
Month 3 — Make one connection	Reach out to one person in a data role. This can be via LinkedIn (request a 15-minute informational conversation), attend one local or virtual data meetup (Meetup.com, dbt Community events, local data engineering groups), or post one data-related observation or project update on LinkedIn. Describe specifically what you will do and what you hope to learn from it. Send a LinkedIn connection request along with a brief message to a Data Quality Analyst or Data Governance expert, ideally located in Texas or working remotely. In the message, I will request a 15-minute informational chat to learn about their journey into the field. If I do not receive a response within a week, I will participate in a free virtual data meetup found on Meetup.com, such as a dbt Community event or a Houston-area data group, and make it a point to introduce myself to at least one attendee. This plan is specific, measurable, and achievable for someone studying in the evenings.

Evening student reality check: Month 1 and 2 work can be done in small sessions — 30–60 minutes a few evenings a week is enough. Month 3 is about one email or one event. None of this requires a large block of free time. The point is consistent, small steps.

What to submit for Part 2

- Three job posting tables from 2a (one table per posting, or all three in a combined table).
- Your skills gap paragraph from 2b.
- Your 90-day plan from 2c — all three months filled in with specific, measurable actions.

Grading Rubric (100 Points)

Specificity is the key criterion. Generic answers that could apply to any dataset or any student earn partial credit at most. Specific, honest answers earn full credit even when the numbers are small or the skills gaps are significant.

Criterion	Full Credit	Partial Credit	Pts
Part 1a–b — Improvements & Value	Two improvements described with before/after; value estimated with reasoning for each; at least one dollar or time figure used.	Two improvements listed; value estimated but reasoning thin or figures absent.	40
Part 1c — Business Case Paragraph	150–250 words; plain language; specific outcome mentioned; at least one quantified estimate; addresses an audience unfamiliar with technical terms.	Present but fewer than 100 words, or heavily technical without business framing.	20
Part 2a — Job Postings	Three real postings; all fields completed; honest assessment of what is and is not already met.	Two postings with most fields; or three postings with gaps in selfassessment columns.	20
Part 2b–c — Gap & 90-Day Plan	Gap identified specifically; all three months have concrete, measurable actions with resource names or platform names.	Gap identified; fewer than three months, or milestones are vague ('learn Python').	20
TOTAL			100